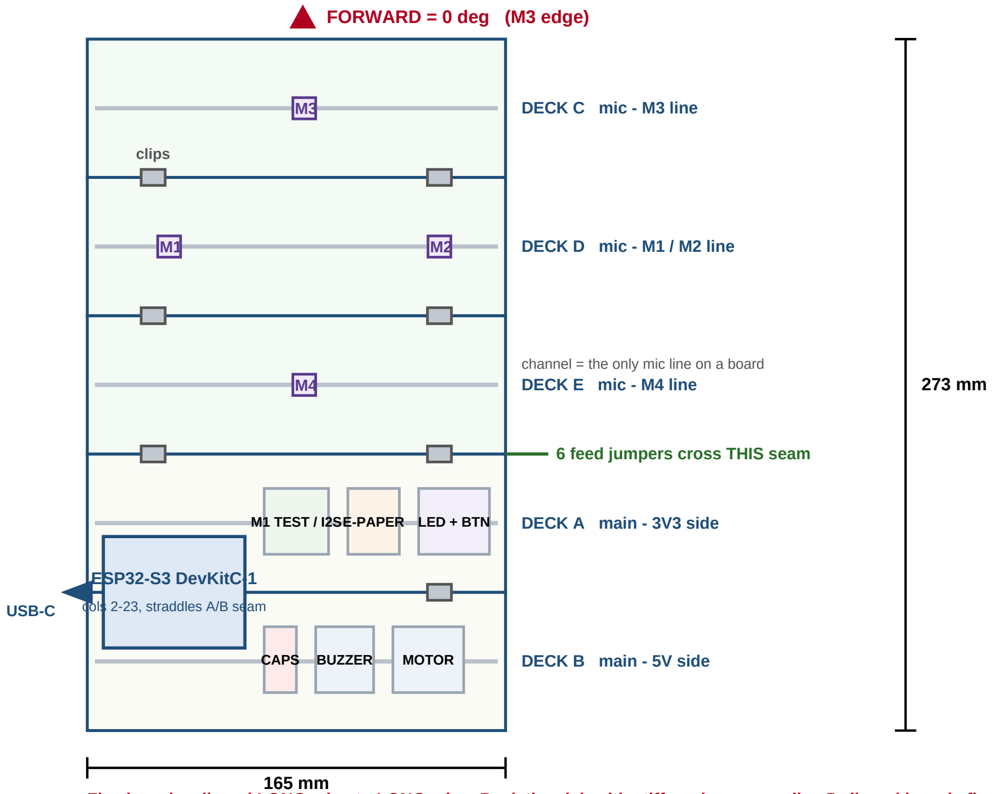
SENTRY-Node - Breadboard Lab Manual v5

v5 - CORRECTED	The INMP441 breakout is a ROUND module with TWO rows of three pads, so it lies flat and STRADDLES a centre channel. One channel per board means one mic line per board, so a 2-D array needs THREE mic boards, not two. Geometry, slab, rail map and assembly all rebuilt around this. New firmware constraint: spatial processing band-limited to 1.5 kHz (§3.1).

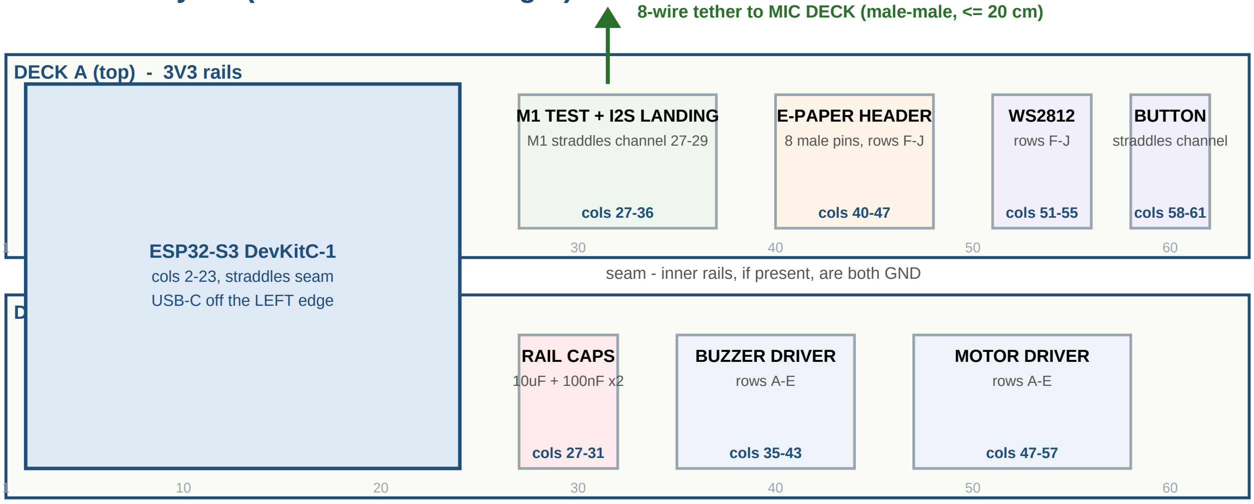
1 · The whole unit



Five boards, clipped LONG edge to LONG edge. Back the slab with stiff card or scrap ally - 5 clipped boards flex.

Stack (top to bottom)	C, D, E (mic) then A, B (main). Five 830-pt boards clipped along LONG edges: 165 x 273 mm.
Why three mic boards	The mic straddles a centre channel, and a board has exactly one channel - so every mic on a board shares the same Y. Two boards can only make a T. Three channel lines give a true cross: M3 / M1+M2 / M4.
Why the baselines are fixed	N-S is channel-to-channel = one board width x2 = 109.22 mm, exact by construction. E-W is 42 column pitches = 106.68 mm, exact by counting. Nothing to measure.
Crossing the E/A seam	Only 6 jumpers: 3V3, GND, SCK, WS, SD_A, SD_B.
Stiffener (recommended)	Five clipped boards flex. Peel the backing and stick the slab to stiff card, thin ply, or scrap aluminium. Flex changes array geometry, which changes bearing.
Vibration tradeoff	Rigid coupling conducts motor buzz into the mics. The motor driver sits on deck B, furthest from the array, and only runs post-detection. If §14 shows the motor-OFF floor changing, unclip the mic deck and rerun on 20 cm jumpers.

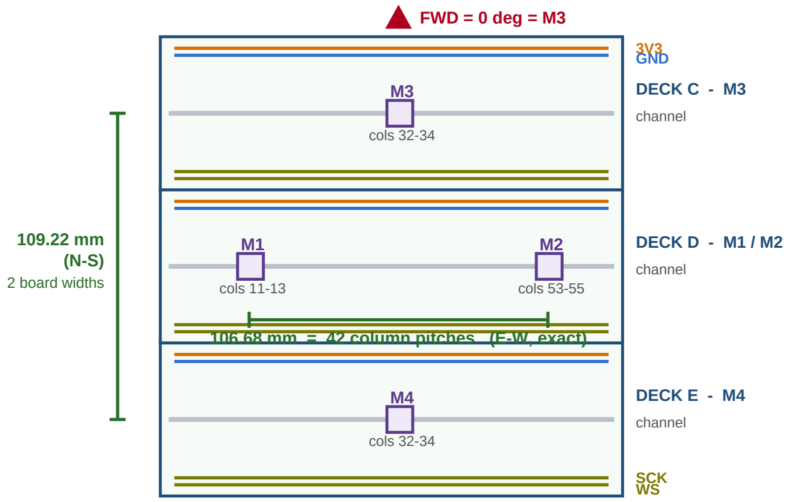
2 · Main deck layout (build order left to right)



Horizontally to scale: 63 columns = one 830-pt board. Keep cols 24-26 empty as a service gap.

Block	Deck	Cols	Rows	Order
DevKitC-1	A+B (straddle)	2-23	all	1
Rail caps	B	27-31	rails	1
M1 test seat + I2S landing	A	27-36	M1 straddles channel at 27-29	2
Button	A	58-61	E/F, straddles channel	3
WS2812	A	51-55	F-J	4
Buzzer driver	B	35-43	A-E	5
Motor driver	B	47-57	A-E	6
E-paper header	A	40-47	F-J	7
Mic deck (separate)	C+D	see §2	see §2	8

3 · Mic deck layout



Each mic STRADDLES its board centre channel and eats 3 columns. One channel per board = one mic line per board.

3.1 · Array coordinates (firmware constants)

Mic	Board	Seat	Pin-1 col	x (m)	y (m)	SD	L/R
M3 North = FWD	C (top)	channel, cols 32-34	32	0.00000	+0.05461	GPIO7	GND
M1 West	D (middle)	channel, cols 11-13	11	-0.05334	0.00000	GPIO6	GND
M2 East	D (middle)	channel, cols 53-55	53	+0.05334	0.00000	GPIO6	3V3
M4 South	E (bottom)	channel, cols 32-34	32	0.00000	-0.05461	GPIO7	3V3
E-W baseline	106.68 mm = 42 column pitches between pin-1s (M1 col 11, M2 col 53). Module centres land on cols 12 and 54, midpoint col 33 = M3/M4 centre column. Exact.						
N-S baseline	109.22 mm = 2 x board width, channel to channel. Exact by construction - no measurement, no per-build constant.						
SPATIAL BAND LIMIT	Aliasing at c/2d: 1570 Hz (N-S), 1608 Hz (E-W), 2247 Hz (diagonals). Band-limit ALL bearing and beamforming to 1.5 kHz. The target fundamental (375-475 Hz) and its first two harmonics sit below this. Detection is per-channel spectral work and is NOT affected - it still uses the full comb.						
Max delay	318 us across the N-S pair - hundreds of samples at 48 kHz, comfortably resolvable.						
Landmark	Coordinates reference the module CENTRE = its middle column. Pin-1 offset is identical on all four, so the geometry holds regardless.						

3.2 · Rail map

THE RULE	Every mic board gets the SAME four rail nets: upper pair = 3V3 / GND, lower pair = SCK / WS. Bridge all three boards at each seam (4 jumpers per seam). SD_A and SD_B are point-to-point - each serves only two mics.

Board	Rail pair	Red strip	Blue strip	Note
C, D, E (mic)	upper	3V3	GND	serves VDD, GND and the L/R strap
C, D, E (mic)	lower	SCK	WS	TAPE-LABEL these - blue is NOT ground here
A (main)	north = E/A seam	3V3	GND	feed point into the mic deck
A (main)	south = A/B seam	3V3	GND	deck A modules; 3V3 decoupling caps
B (main)	north = A/B seam	3V3	GND	buzzer supply, just across the seam
B (main)	south = slab bottom	5V	GND	motor supply; 5V caps. ONLY 5V strip in the build.

3.3 · Per-mic jumper list (6 per mic, 24 total)

Pad	Goes to	Tap from	Reach
VDD	3V3 (upper red)	free holes of its own column, upper half	~13 mm
GND	GND (upper blue)	same	~13 mm
SCK	SCK (lower red)	free holes of its own column, lower half	~13 mm
WS	WS (lower blue)	same	~13 mm
L/R	GND for M1 and M3, 3V3 for M2 and M4	cross to the upper pair	~40 mm
SD	GPIO6 for M1+M2, GPIO7 for M3+M4	point-to-point: feed to M1 then daisy to M2; feed to M4 then daisy to M3	up to 160 mm - use 20 cm jumpers

4 · Master pin map

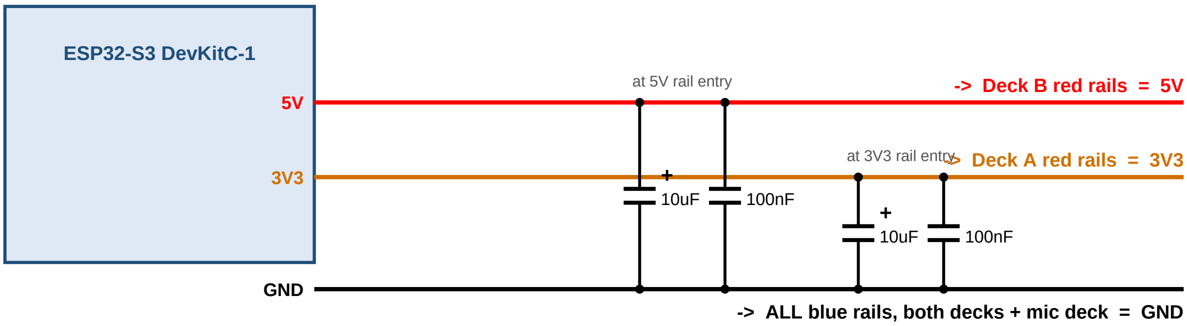
Signal	GPIO	Colour	Notes
I2S SCK - all 4 mics	4	yellow	shared
I2S WS - all 4 mics	5	white	shared
I2S SD bus A - M1, M2	6	blue	M1 left slot, M2 right slot
I2S SD bus B - M3, M4	7	green	2nd I2S peripheral; silent until beamforming
E-paper CS/DIN/CLK/DC/RST/BUSY	10/11/12/13/14/15	purple	wire by HAT labels only
WS2812 data	16	grey	via 330R
Buzzer base	17	grey	via 1k
Motor base	18	grey	via 1k
Snooze button	21	grey	to GND, internal pull-up
KEEP CLEAR	0, 3, 19, 20, 26-37, 43-46	-	strapping / USB / flash / PSRAM / UART
Spares	1, 2, 8, 9, 39-42, 47	-	38 / 48 may drive the onboard RGB (board revision dependent)

5 · Mechanical conventions

Header soldering	Mic: two SEPARATE 3-pin strips, one per pad row. Plastic spacer flush to the underside; LONG pins point DOWN into the breadboard; solder the SHORT ends on the labelled top face. Keep solder and flux clear of the centre port hole. Same long-pins-down rule for the other modules.
Mic orientation	Module lies FLAT, pins down, straddling the channel. The acoustic port is the small hole at the centre of the top face - it faces UP and must never be taped, glued, or covered. Rotate all four identically (silkscreen readable from the M3 / forward edge) so wiring and responses stay matched.
Mic is MONO	One transducer, one channel. L/R selects which half of the I2S stereo frame it drives, so two mics share one SD wire. 4 mics = 4 channels; all four feed beamforming, orthogonal pairs are just the bearing readout.
Mic is OMNI	The INMP441 hears all directions equally - at 150 Hz-2 kHz the wavelength is 17-90 cm against a ~1 mm port, so there is no usable directivity in the capsule. Direction comes ONLY from arrival-time differences between separated mics. Facing all four the same way is therefore required, not merely tidy: it keeps the four channel responses matched, which is what the TDOA maths assumes. Rotating one to "cover the front" gains no coverage and adds phase mismatch.
Seating - two-row parts	The mic has TWO rows of three pads, so it must straddle the centre channel exactly like a DIP chip: 3 columns wide, one pad row in the F-J half, the other in the A-E half. Six independent nets, tapped from the free holes of each column. The tact switch straddles for the same reason.
One line per board	A board has exactly one centre channel, so every mic on it shares the same Y. This is why the array needs three mic boards - it is a hard mechanical constraint, not a preference.
Jumper gender	DevKit, tether, mic deck, all modules: male-male throughout (every connection is breadboard-hole to breadboard-hole). E-paper HAT uses its own supplied cable.
Dressing	SCK/WS/SD as one short bundle twisted with a spare GND; keep clear of motor leads. Power stays on rails. Trim excess; no loops.

6 · Power rails [Deck A+B, cols 2-31]

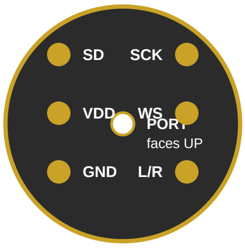
USB-C from Mac = only power source



Parts	Checks (gate - nothing else is wired until these pass)
10uF electrolytic x2, 100nF ceramic x2, 4x male-male jumpers, 2 board clips	3V3 rail = 3.25-3.40 V • 5V rail = 4.6-5.15 V • GND continuity deck A rail to deck B rail < 1 ohm • electrolytic stripe = GND side

7 · Mic M1 - first light (= Stage 1b) [Deck A: M1 straddles the channel at cols 27-29; jumper landing cols 30-36]

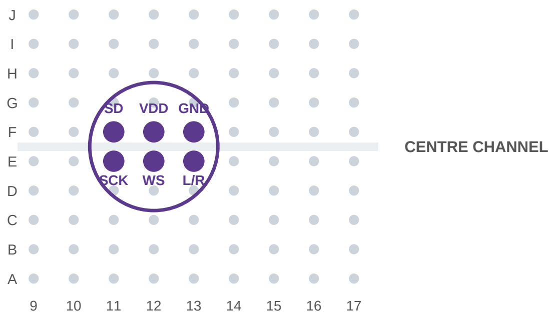
TOP VIEW (as it lies on the board)



Two rows of three pads - identify by silkscreen on YOUR module; variants differ.

SEATING - straddles the channel, 3 columns wide

One pad row in the F-J half, the other in the A-E half.



column numbers (M1 shown: pin-1 col 11)

Tap each net from the free holes of its own column:
rows G-J for the upper pads, rows A-D for the lower pads.

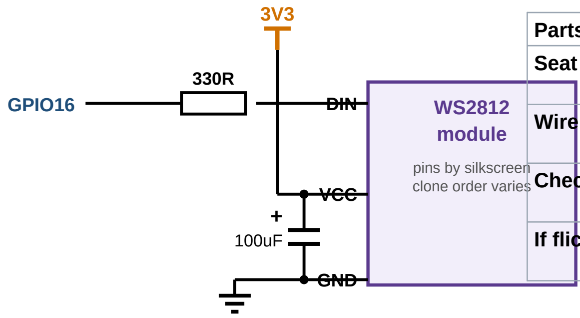
Seat	M1 straddles deck A centre channel at cols 27-29, port up. Deck A has its own channel - use it for this test; M1 moves to deck D later.
Wiring	VDD->3V3 rail • GND->GND rail • SCK->GPIO4 • WS->GPIO5 • SD->GPIO6 • L/R->GND. Tap each from the free holes of its own column.
Checks	Live 24-bit stream at the configured rate • clap = clean full-scale transient • no stuck or repeated words • steady noise floor
Then	M1 moves to the mic deck at §13. Leave this strip in place - it becomes the tether landing.

8 · Snooze button [Deck A, cols 58-61, straddling the centre channel]

Use DIAGONAL pins - the two pins on one side are always shorted together internally.

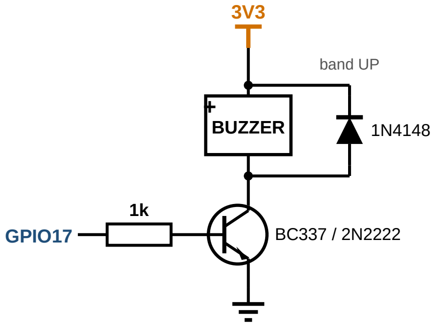
Part	6 mm 4-pin tact switch (through-hole), legs straddle the channel: two legs in row E, two in row F
Seat	Legs into cols 58 + 60, rows E and F. Straddling the channel is what keeps the two sides isolated.
Wire	GPIO21 -> col 58 (row G) • col 60 (row D) -> GND rail. Use one leg from each DIAGONAL pair.
Optional	100nF across the same two columns if it double-fires
Check	One clean press/release logged per press; pressed reads LOW

9 · WS2812 status LED [Deck A, cols 51-55, rows F-J]



Parts	WS2812 breakout, 330R, 100uF electrolytic
Seat	Module pins into cols 51-55, row F. Identify DIN / VCC / GND by SILKSCREEN - clone pin order varies.
Wire	GPIO16 -> 330R -> DIN • VCC -> 3V3 rail • GND -> GND rail • 100uF across VCC-GND at the module, + to VCC
Check	Commanded R, then G, then B display as those colours (catches colour order and the 3V3 data threshold at once)
If flicker	Swap to another module from the 5-pack first. Changing the supply is a design change - flag it.

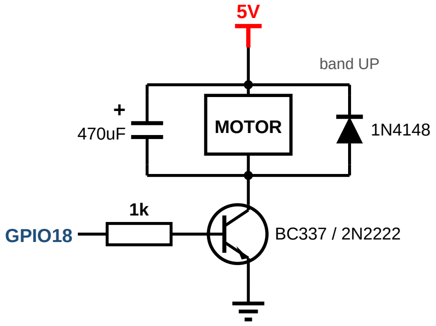
10 · Buzzer driver [Deck B, cols 35-43, rows A-E]



GPIO HIGH = on. Load between supply and collector, emitter to GND. Check pinout: BC337 and 2N2222 differ.

Part ID	Black cylinder, two stiff pins, "+" sticker (3 V active buzzer)
Parts	Buzzer, BC337 or 2N2222, 1k, 1N4148
Seat	Buzzer cols 35/36 • NPN cols 38-40 • 1k cols 41-43
Wire	Buzzer + -> 3V3 rail • buzzer - -> NPN collector • emitter -> GND • GPIO17 -> 1k -> base • 1N4148 across buzzer, band to 3V3
Check	Beeps on a test pulse • ~0.7 V base-emitter when on • silent and cool when off

11 · Motor driver [Deck B, cols 47-57, rows A-E]



GPIO HIGH = on. Load between supply and collector, emitter to GND. Check pinout: BC337 and 2N2222 differ.

Part ID	Coin / pager disc, red and blue flying leads (solder to male pins first)
Parts	Motor, BC337 or 2N2222, 1k, 1N4148, 470uF
Seat	Motor leads cols 47/48 • NPN cols 50-52 • 1k cols 53-55 • 470uF cols 56/57 to rails
Wire	Identical to §10 but supply = 5V rail and GPIO18. Motor leads go either way round. Band of the 1N4148 to 5V.
Check	Spins on a test pulse • with the motor OFF, M1 noise floor unchanged from §7 (the motor must not contaminate audio)

12 · E-paper display [Deck A, cols 40-47, rows F-J]

HAT pin	VCC	GND	DIN	CLK	CS	DC	RST	BUSY
Col	40	41	42	43	44	45	46	47
Goes to	3V3	GND	GPIO11	GPIO12	GPIO10	GPIO13	GPIO14	GPIO15

Seat	8-pin male strip into cols 40-47 row F. HAT hangs off-board on its supplied cable, lying flat right of deck A.
Rule	Wire by the labels on the HAT and cable heat-shrink, NEVER by wire colour - Waveshare colour order varies by batch. No extra passives.
Check	Demo refresh draws • BUSY toggles during refresh • M1 noise floor unchanged while refreshing

13 · Full mic array [mic decks C, D, E - see §3]

13a	Before anything: measure the pad-row spacing on one module. It must equal the channel width (0.3 in / 7.62 mm) to straddle. If it is 0.2 in, stop and flag - the module needs short link wires instead of a direct seat.
13b	Clip C to D to E, then clip E onto the top long edge of deck A. Back the whole slab with stiff card or ally.
13c	Set the rails per §3.2 on all three mic boards: upper pair 3V3 / GND, lower pair SCK / WS. Tape-label the lower pairs - blue is not ground there. Bridge all three boards at both seams (4 jumpers per seam).
13d	Seat the mics straddling each channel: M3 cols 32-34 on C, M1 cols 11-13 and M2 cols 53-55 on D, M4 cols 32-34 on E. All four rotated identically, ports up and clear.
13e	Wire the 6 jumpers per mic (§3.3). L/R: GND on M1 and M3, 3V3 on M2 and M4.
13f	Six feed jumpers across the E/A seam: 3V3 and GND from deck A north pair; SCK and WS from the §7 I2S strip to deck E lower pair; SD_A to M1 column; SD_B to M4 column. M1 leaves deck A.
13g	Set the firmware coordinates from §3.1 and enable the 1.5 kHz spatial band limit before running any bearing test.

Tap test (mandatory before any beamforming): with 4-channel capture running, tap each mic in turn - the spike must land on its expected channel (M1 bus A left, M2 bus A right, M3 bus B left, M4 bus B right). Then one sharp clap: all four channels show it within ~1 ms.

14 · Full smoke test

Run	4-channel capture + LED activity + e-paper refresh + commanded buzzer-and-motor alert burst, all at once
Pass	Noise floor unchanged with LED and e-paper active • buzzer/motor mask audio only during the alert • floor returns to baseline after

15 · Parts

From your order	Qty	From York lab stock	Qty
ESP32-S3 DevKitC-1 N16R8	1 (+1 spare)	830-pt breadboards	4 (2 main + 2 mic deck)
INMP441 modules	4	Board clips / spare 830-pt	as needed
Waveshare 2.13 e-paper HAT V4 + cable	1	Jumpers male-male	60+
WS2812 breakout	1 of 5	BC337 or 2N2222	3
Active buzzer 3 V	1 of 10	1N4148	4
Vibration motor 3-5 V	1 of 10	1k / 330R resistors	3 / 2
USB-C DATA cable	1	100nF / 10uF / 100uF / 470uF	8 / 3 / 2 / 1
NOT used: slide switch, LiPo gear, power bank	0	Tact buttons, 40-pin male header, stiff card or ally backing, DMM, marker	2 / 2 / 1 / 1 / 1

16 · Risk flags

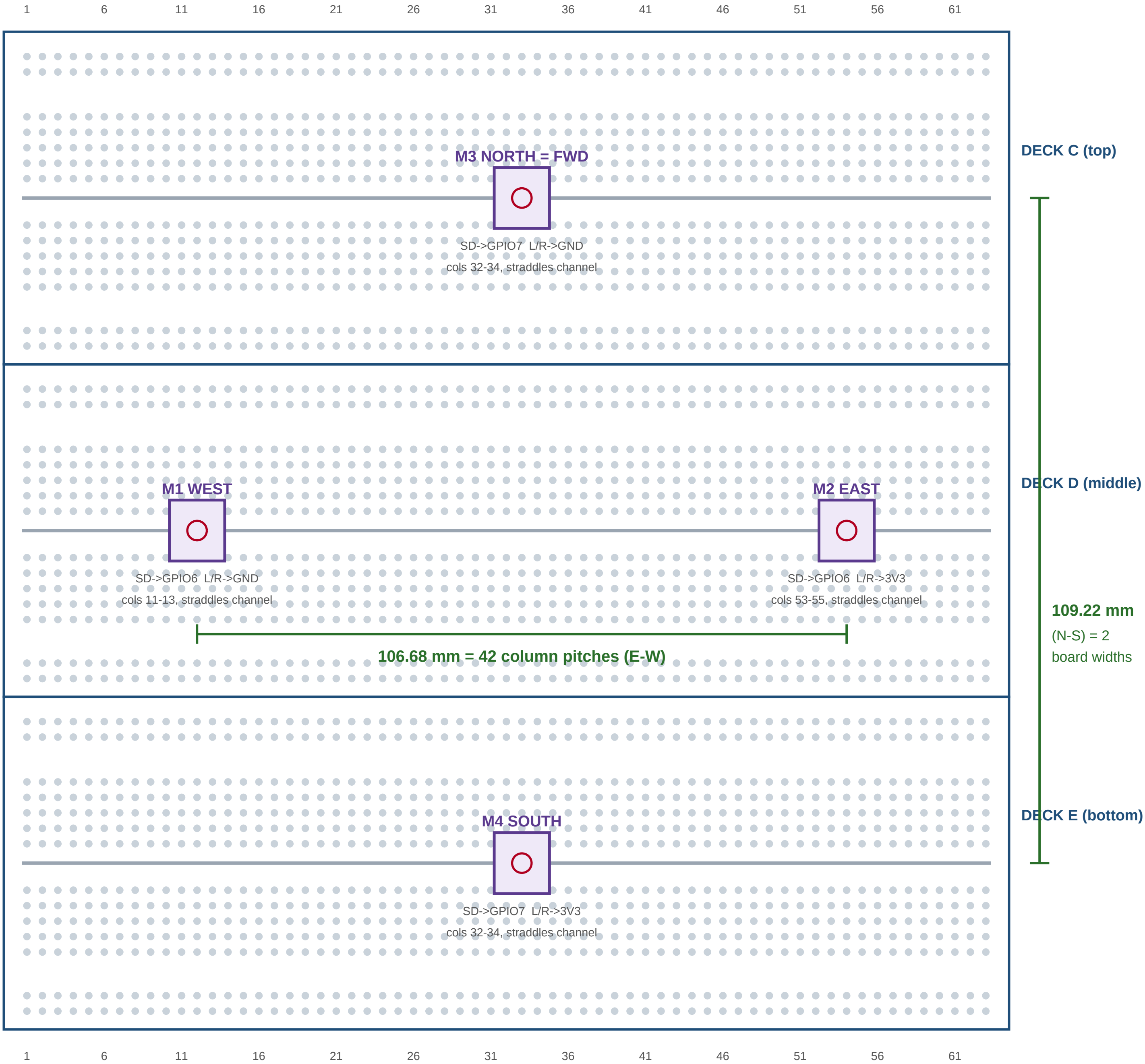
F1	INMP441 quantity: cart line shows 3. If those are single modules you are one short of the array. Verify pack size / reorder before lab day.
F2	Measure the pad-row spacing before lab day. 0.3 in straddles the channel directly; 0.2 in does not and needs link wires. This gates the whole array build.
F3	WS2812 at 3V3 data: usually fine, some clones need >3.5 V. Symptom = flicker / wrong colour. Swap modules first.
F4	DevKitC-1 barely fits: straddle cols 2-23. Single-board fallback works but is cramped.
F5	E-paper: wire by HAT/cable labels only - Waveshare colours are not consistent between batches.
F6	Board dimensions vary slightly by manufacturer: confirm the N-S baseline by measurement, not by trusting 83.82 mm.
F7	USB current: use the Mac port or a powered hub. Motor pulses peak near 0.4 A.
F8	Slab rigidity: five clipped boards flex, and flex moves the array geometry. Back the slab, carry it flat, re-run the tap test after transport.
F9	Port hole faces up on every module - do not rest anything on the mic deck, and keep the backing material off the top face.

17 · Change-log entries for the Master Brief (§14)

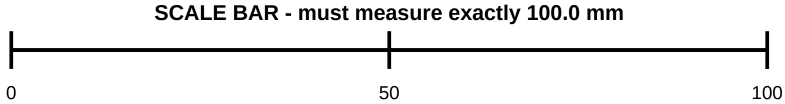
- CORRECTION: the INMP441 breakout in use is the round 2 x 3-pad variant. It lies flat and straddles a breadboard centre channel; it does not stand in a single row. Earlier plate / riser / vertical-seating notes are void.
- A board has one centre channel, so it can host only one mic line. A 2-D array therefore requires THREE mic boards. Mic deck is now decks C, D, E.
- Array geometry fixed by construction: symmetric cross, E-W 106.68 mm (42 column pitches), N-S 109.22 mm (2 board widths). No per-build measurement, no MIC_SPACING constant - four explicit (x, y) coordinates.
- NEW FIRMWARE CONSTRAINT: spatial aliasing at ~1.57 kHz, so all bearing and beamforming is band-limited to 1.5 kHz. Target fundamental and first two harmonics sit below it. Detection is unaffected.
- Breadboard v3: single five-board slab (C, D, E, A, B), 165 x 273 mm, backing recommended. Slide switch still deleted; build remains USB-powered.
- Confirmed: the mic is mono AND omnidirectional. L/R is the I2S frame slot, not a channel and not a direction; bearing comes solely from inter-mic arrival time.

MIC DECK MAP - 1:1

Print at 100% / "Actual size", then lay your three clipped boards on top.



SCALE BAR - must measure exactly 100.0 mm



1 - Red circle = the module centre column. Coordinates in section 3.1 reference it.

2 - Each mic lies FLAT and straddles its board centre channel, three columns wide, port hole facing UP.

3 - Both baselines are exact by construction: nothing on this map needs measuring, only checking.

4 - Run the tap test (section 13) before trusting any bearing.